Thesis Talk

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May 4, 2026

Overview of Topics

- Context
- Terminology
- Overview of Models
- Results of Cohesive Zone Model
- Disucussion
- Outlook

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- At the beginning, oversample and knit an entire top because the run time was too long
- Trying to make an accurate model for crack propagation in snow using SALVUS (FE)
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will propably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:

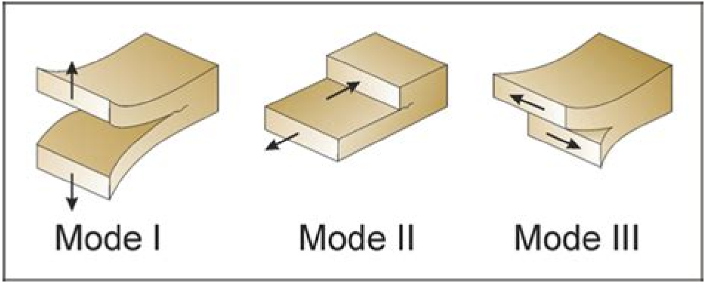

Figure: Fracture modes according to Philipp et al., 2013

Words I will propably say a lot explained II

- Anticrack: weak layer collapses under loading, slab starts bending, collapse and cracks caused by bending propagate (Gaume et al., 2018)
- Sub-Rayleigh: Cracks that propagate at speeds lower than the shear wave speed in snow (Trottet et al., 2022)
- Supershear: Cracks propagate faster than shear wave speed in snow (Trottet et al., 2022)

Overview of Models

- Model domain 400m wide, 3m high
- Absorbing boundary conditions 30m thick along top and both sides of domain
- No ABC at bottom because of theoretical snow interface
- 1.5m snow layer, 1.5m air layer
- Multiple source versions: single source, source line with sources firing with a time delay (moving source) and rupture front
- Sources buried in snow (like crack)
- For moving source: sub-rayleigh and supershear speeds
- 10Hz central frequency ricker source wavelet

My latest creation: the cohesive zone model (CZM)

- Model based on the cohesive zone model by Mahajan and Joshi, 2008
- In CZM, fracturing is a gradual process, separation of the crack faces takes place over an extended crack tip, the cohesive zone (Mahajan & Joshi, 2008)
- Resistance to opening by cohesive traction (Mahajan & Joshi, 2008)
- In other words: there is a zone ahead of the crack tip, which already has some damage of the bonds
- Cracks represented as a fraction between different crack modes (Mode I-III)
- Caused by a a fault zone: starts at x=30m and ends at x=270m, 0.5m depth (middle of snow layer)
- Sources every 10cm firing with a time delay (dependend on supershear vs sub-rayleigh)

Supershear vs Sub-Rayleigh Space-Time Plots - Velocity

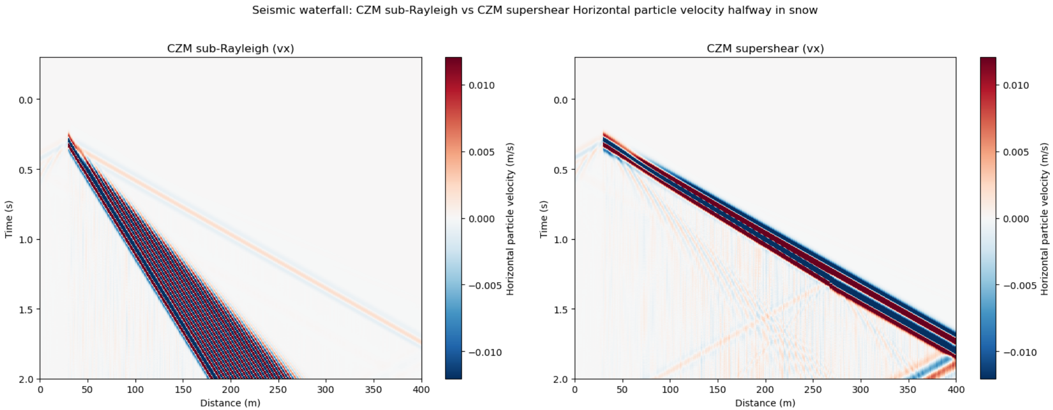

Figure: Horizontal Particle Velocity (vx) in sub-rayleigh (left) and supershear case (right)

Supershear vs Sub-Rayleigh Space-Time Plots - Strain

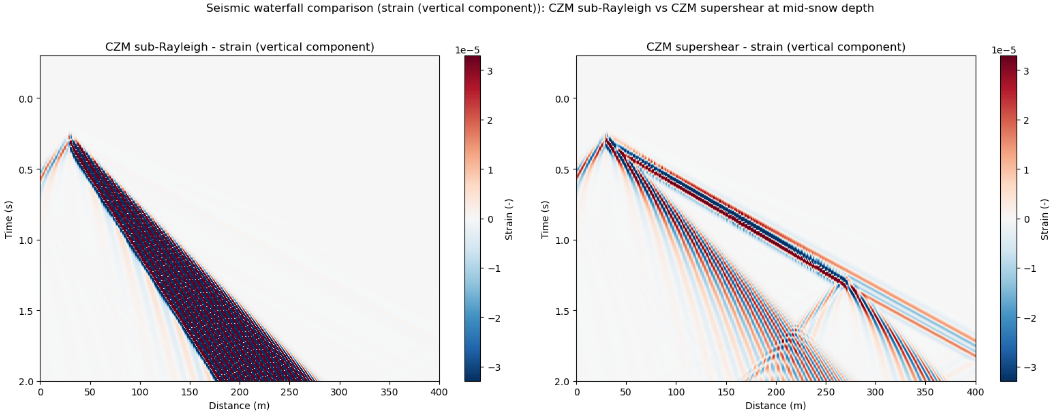

Figure: Strain in sub-rayleigh (left) and supershear case (right)

Possible interpretations of space-time plots I

- Sub-Rayleigh and Supershear case look very different (yay!)
- Blue-red-blue arrival before the main wavefront could be flexural wave (bending of the slab near the source)
- Blue first: negative velocity –> source in negative y (slab is pushed down by the source)
- Red next: because of elastic rebound of slab
- Blue again: oscillation in source direction after rebound
- This can be seen in vx for sub-rayleigh and strain for supershear
- In sub-rayleigh, waves travel slower than the flexural waves, which is very noticeable when looking at vx
- In supershear vx, rupture arrives with flexural waves
- Flexural waves can be seen in strain supershear plot: because strain is spatial derivative, which acts as a high-pass and amplifies flexural waves
- Supershear case: Strong first arrivals, weak arrivals at later times
- "Reflection" seen in bottom rigth corner is rupture arrest front (energy could increase because rupture stops at a point)

Bibliography I

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